# SOFTWARE ARCHITECTURE

## Decentralized Architecture

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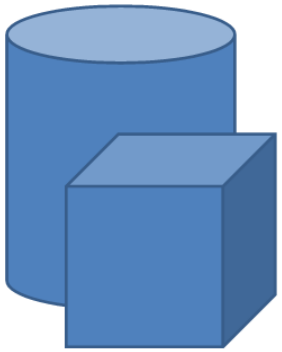
# Content

- Decentralized architecture
- Cryptocurrency
- Blockchain
- Wallet

# Centralized, decentralized processing

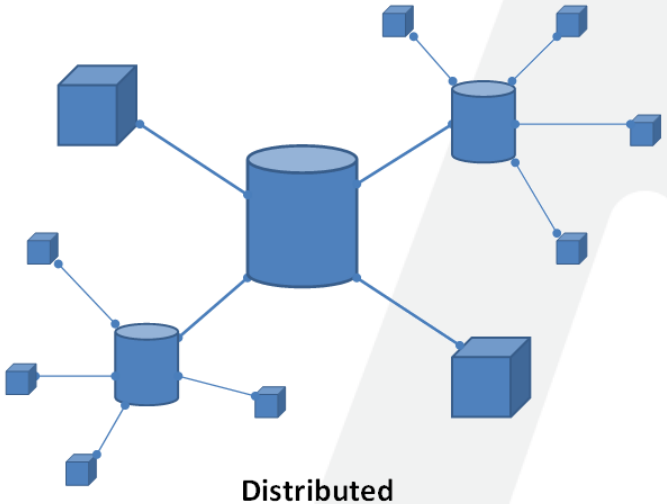
- In **centralized** processing all the terminals are controlled by a **single processor** (CPU) and any command can be fulfilled by a single processor and this type of network is called centralized network.
- In **decentralized** processing, there are **different CPU connected on the network** and each processor can do its job independent of each other.

# Centralized, distributed, and decentralized



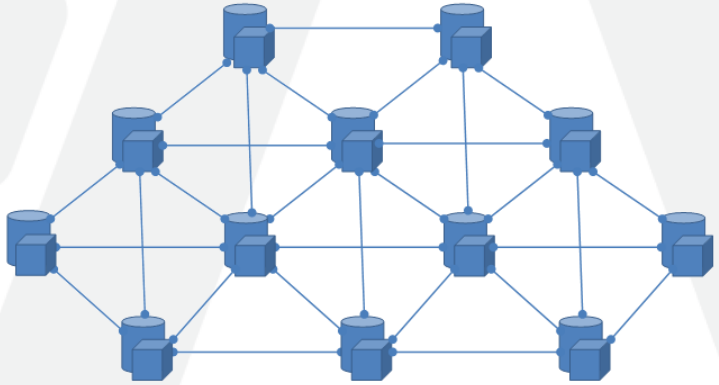
**Centralized**

*one node does everything*



**Distributed**

*nodes distribute work to sub-nodes*



**Decentralized**

*nodes are only connected to peers*

# These things are decentralized

- Access
- Organizational governance
- Computation
- Financial Instruments
- Wealth

# What is decentralized computing?

- Computing is decentralized when critical application services are carried out by individual computing devices or nodes on a distributed network, with no central location.
- If you can't point to a single service address and disable it to shut down core application functionality for all users, chances are good that you have a decentralized computing architecture.

# Decentralized Applications (DApps)

- **Cryptocurrency transactions** — the foundation of the internet of value.
- **Cryptocurrency exchange** — enable smart wallet meta-transactions to pay protocol token fees and blockchain gas fees, so users can take advantage of decentralized services which need payments.
- **User wallet-secured authentication and authorization**. No more passwords to remember or lose.
- **Decentralized identity** (often represented by a wallet public address). With DApps, your identity follows you effortlessly between applications.
- **Decentralized user profile data**. Grant, revoke, or reject application access to your user profile data.

# Decentralized Applications (DApps)

- **Decentralized user content storage.** Ditto, but for all your content — posts, videos, photos, and comments.
- **Decentralized networking and compute services**. User-owned mesh networking, gig economy style sharing of your computer's storage and CPU resources.
- **Cryptographic hashing and signatures**. Proof of existence, decentralized authorization, proof of resource ownership, permission, or control. These are basic primitives that are required for decentralized application security to work.
- **Smart contracts**. Programmable money transactions and agreements, enforced by code, so there's no need to trust participants.

# What is Cryptocurrency?

- A cryptocurrency is not a type of currency that can be used in the real world.
- It can be used to perform transactions only in the digital world.
- In order to buy/sell using a cryptocurrency, it must be converted from a digital form to some existing currency that is used in the real world. For example, Dollars, Rupees, etc.
- Cryptocurrencies don't have a central issuing authority instead using a decentralized system to record transactions and issue new units.

# What is Cryptocurrency?

- Cryptocurrency is a digital payment system that does not rely on banks to verify transactions.
- Cryptocurrency payments exist purely as digital entries to an online database. When cryptocurrency funds are transferred, the transactions are recorded in a public ledger.

# Cryptocurrency characteristics

- In cryptocurrency, “coins” (which are publicly agreed on records of ownership) are generated or produced by “miners”.
- These miners are people who run programs on ASIC (Application Specific Integrated Circuit) devices made specifically to solve proof-of-work puzzles.
- The work behind mining coins gives them value, while the scarcity of coins and demand for them causes their value to fluctuate.
- Cryptocurrencies can be used for buying goods just like fiat currency.
- Cryptocurrencies use encryption to verify and protect transactions.
- It does not exist in physical form and is not typically issued by any central authority.
- They use decentralized control in contrast to central bank digital currency.

# Cryptocurrency Examples (1/2)

- Bitcoin: Bitcoin is the most widely accepted cryptocurrency. Founded in 2009 by Satoshi Nakamoto, it is still the most traded. It is a decentralized digital currency that can be transferred on a peer-to-peer bitcoin network.
- Ether: Ether is the native cryptocurrency of the Ethereum blockchain network. Each Ethereum account has an ETH balance and may send ETH to any other account. The smallest subunit of Ether is known as Wei.
- Litecoin: Litecoin is a peer-to-peer cryptocurrency and in technical terms, Litecoin is nearly identical to Bitcoin. It uses scrypt in its proof-of-work algorithm. It is an adaptation of Bitcoin that is intended to make payment easier.

# Cryptocurrency Examples (2/2)

- Stablecoins: These are the class of cryptocurrencies whose values are designed to stay stable relative to real-world assets like the U.S. Dollar.
- Solana: Solana is a competitor of Ethereum whose main emphasis is on speed and cost-effectiveness.

# Cryptocurrency vs Fiat Currency

- Two things that make cryptocurrency working and fiat currency working different:
  - Transactions
  - Consensus protocol

# Transactions

- The transactions performed in the crypto world are very different than those that of which are performed in the real world. Let's consider that Alice wants to buy a Bicycle.
  - **Real-world**: In the real-world Alice can pay in any available currency. The seller will return the change if any to Alice.
  - **Crypto world**: Suppose the bicycle costs 0.6 BTC and Alice has 0.7 BTC in the Bitcoin Wallet. Alice must consider the whole amount i.e., 0.7 BTC
    - Transaction 1: Transfer only 0.6 BTC from Bitcoin wallet to the seller's wallet. Now, Alice has already exhausted 0.6 out of 0.7 BTC. The remaining 0.1 BTC must be transferred back to Alice's wallet. There is no change in BTC being offered by the seller to Alice.
    - Transaction 2: Alice offers 0.1 BTC back to herself. So, 0.1 BTC is an unspent transaction amount in Alice's wallet.

# Consensus protocol

- Consensus decision-making is a group decision-making process in which group members develop and agree to support a decision in the best interest of the whole.
- Basically, it states that the longest valid chain in the Blockchain network should exist on every node in the Network.

# How Does Cryptocurrency Works?

- Cryptocurrencies are not regulated or controlled by any central authority hence cryptocurrency works outside the banking system using different types of coins.
  - **Mining**: Cryptocurrencies are generated through the process called Mining. In this process, the miners are required to solve a mathematical puzzle over a specially equipped computer system to be rewarded with bitcoins in exchange.
  - **Buying**, selling, and storing: Users can buy cryptocurrencies from central exchanges, brokers, or individual currency owners and sell crypto to them. Cryptocurrencies can be stored in wallets.
  - **Investing**: Cryptocurrencies can be transferred from one digital wallet to another. Cryptocurrencies can be used for the following purposes:
    - Buying goods and services.
    - Trade-in them.
    - Exchange them for cash.

# How To Store Cryptocurrency (1/2)

- Once the cryptocurrency is purchased, it needs to be stored safely to protect it from hackers.
  - **Custodial Wallet**:
    - A third party such as a crypto exchange stores the cryptocurrency either through cold storage or hot storage, or a combination of the two.
    - This is the simplest and most convenient method for the users as it requires less work on the user part.
  - **Cold Wallet**:
    - These are also known as Hardware wallets. It is an offline wallet in which hardware connects to the computer and stores the cryptocurrency.
    - The device connects to the internet at the time of sending and receiving cryptocurrency but other than that the cryptos are safely stored offline.

# How To Store Cryptocurrency (2/2)

- Once the cryptocurrency is purchased, it needs to be stored safely to protect it from hackers.
  - **Hot Wallet**:
    - These are the applications that store cryptocurrencies online.
    - These are available as desktop or mobile apps.
  - **Paper Wallet**:
    - This is also known as a physical wallet.
    - It is a printout of the public and private keys available as a string of characters or scannable QR codes. To send crypto scan the public and private keys and crypto will be received using the public keys.

# What Can You Do With Cryptocurrency

- Here are some of the examples:
  - **Shopping**: Some luxury retailers like Rolex and Patek Philippe accept cryptocurrency as a form of payment.
  - **Insurance**: Some insurance companies like Premier Shield insurance accept Bitcoin for premium payments.
  - **Gift**: Cryptocurrency can be a great gift for persons who want to learn and invest in new technology.
  - **Travel**: As crypto is not tied to a specific country, thus traveling with crypto can save a lot on money exchange fees.

# Advantages of Cryptocurrencies

- **Private and Secure**: Blockchain technology ensures user anonymity and at the same time the use of cryptography in blockchain makes the network secure for working with cryptocurrencies.
- **Decentralized**, Immutable, and Transparent: The entire blockchain network works on the principle of shared ownership where there is no single regulating authority and the data is available to all the permissioned members on the network and is tamper-proof.
- **Inflation Hedge**: Cryptocurrencies are a good means of investing in times of inflation as they are limited in supply and there is a cap on mining any type of cryptocurrency.
- **Faster Settlement**: Payments for most cryptocurrencies settle in seconds or minutes. Wire transfers at banks can cost more and often take three to five business days to settle.
- **Easy Transactions**: Crypto transactions can be done more easily, in a private manner in comparison to bank transactions. using a simple smartphone and a cryptocurrency wallet, anyone can send or receive a variety of cryptocurrencies.

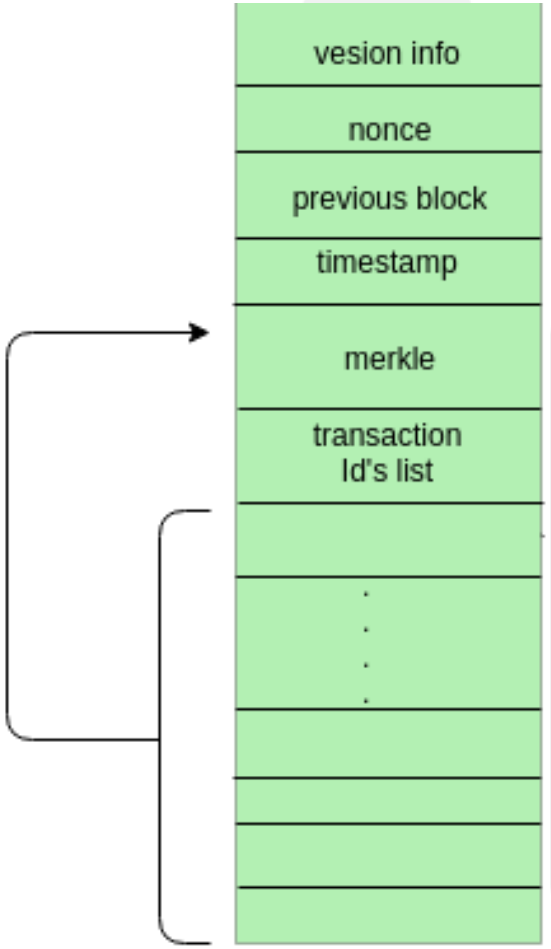
# Disadvantages of Cryptocurrencies

- **Cybersecurity issues**: Cryptocurrencies will be subject to cyber security breaches and may fall into the hands of hackers. Mitigating this will require continuous maintenance of security infrastructure.
- **Price Volatility**: Cryptocurrencies are highly volatile in terms of price as they have no underlying value and there is a supply-demand-like equation that is used to determine the price of cryptocurrencies.
- **Scalability**: Scalability is one of the major concerns with cryptocurrencies. Digital coins and tokens adoption is increasing rapidly but owing to the sluggish nature of the blockchain makes cryptocurrencies prone to transaction delays. Cryptocurrencies cannot compete with the number of transactions that payment giants like VISA, and Mastercard processes in a day.
- **Less awareness:** Cryptocurrency is still a new concept for the people and the long-term sustainability of cryptocurrencies remains to be seen.

# Blockchain

- Blockchain could be a data structure that could be a growing list of information blocks.
- The knowledge blocks area unit coupled along, such recent blocks can't be removed or altered. Blockchain is the backbone Technology of Digital Cryptocurrency Bitcoin.
- The blockchain is a distributed database of records of all transactions or digital event that have been executed and shared among participating parties.
- Each transaction verified by the majority of participants of the system. It contains every single record of each transaction.

# A block in a Blockchain



# How does Blockchain Technology work?

- **Distributed Database**:
  - There is no Central Server or System which keeps the data of the Blockchain. The data is distributed over Millions of Computers around the world
  - This system allows Notarization of Data as it is present on every Node and is publicly
- **A network of nodes**:
  - A node is a computer connected to the Blockchain Network. Node gets connected with Blockchain using the client. Client helps in validating and propagating transaction on to the Blockchain.
  - When a computer connects to the Blockchain, a copy of the Blockchain data gets downloaded into the system and the node comes in sync with the latest block of data on Blockchain.
  - The Node connected to the Blockchain which helps in the execution of a Transaction in return for an incentive is called Miners.

# Building trust with Blockchain

- **Distributed**: The distributed ledger is shared and updated with every incoming transaction among the nodes connected to the Blockchain. All this is done in real-time as there is no central server controlling the data.
- **Secure**: There is no unauthorized access to Blockchain made possible through Permissions and Cryptography.
- **Transparent**: Because every node or participant in Blockchain has a copy of the Blockchain data, they have access to all transaction data. They themselves can verify the identities without the need for mediators.
- **Consensus-based**: All relevant network participants must agree that a transaction is valid. This is achieved using consensus algorithms.
- **Flexible**: Smart Contracts which are executed based on certain conditions can be written into the platform. Blockchain Network can evolve in pace with business processes.

# Benefits of Blockchain Technology

- **Time-saving**: No central Authority verification needed for settlements making the process faster and cheaper.
- **Cost-saving**: A Blockchain network reduces expenses in several ways. No need for third-party verification. Participants can share assets directly. Intermediaries are reduced. Transaction efforts are minimized as every participant has a copy of shared ledger.
- **Tighter security**: No one can temper with Blockchain Data as it shared among millions of Participant. The system is safe against cybercrimes and Fraud.
- **Collaboration**: It permits every party to interact directly with one another while not requiring third party negotiate.
- **Reliability**: Blockchain certifies and verifies identities of every interested party. This removes double record, reducing rates and accelerate transactions.

# Application of Blockchain (1/3)

- **Leading Investment Banking Companies** like Credit Suisse, JP Morgan Chase, Goldman Sachs and Citigroup have invested in Blockchain and are experimenting to improve the banking experience and secure it.
- **Following the Banking Sector**, the Accountants are following the same path. Accountancy involves extensive data, including financial statements spreadsheets containing lots of personal and institutional data. Therefore, accounting can be layered with blockchain to easily track confidential and sensitive data and reduce human error and fraud. Industry Experts from Deloitte, PwC, KPMG and EY are proficiently working and using blockchain-based software.
- **Booking a Flight** requires sensitive data ranging from the passenger's name, credit card numbers, immigration details, identification, destinations, and sometimes even accommodation and travel information. So the sensitive data can be secured using blockchain technology. Russian Airlines are working towards the same.

# Application of Blockchain (2/3)

- Various industries, including **hotel services**, pay a significant amount ranging from 18-22% of their revenue to third-party agencies. Winding Tree works extensively with Lufthansa, Air France, Air Canada, and Etihad Airways to cut short third-party operators charging high fees.
- **Barclays** uses Blockchain to streamline the Know Your Customer (KYC) and Fund Transfer processes while filling patents against these features.
- **Visa** uses Blockchain to deal with business-to-business payment services.
- **Universes** Blockchain to track all their transactions in the supply chain and maintain the product's quality at every stage of the process.

# Application of Blockchain (3/3)

- **Walmart** has been using Blockchain Technology for quite some time to keep track of their food items coming right from farmers to the customer. They let the customer check the product's history right from its origin.
- **DHL** and **Accenture** working together to track the origin of medicine until it reaches the consumer.
- **Pfizer**, an industry leader, has developed a blockchain system to keep track of and manage the inventory of medicines.
- **The government of Dubai** looking forward to making Dubai the first-ever city to rely on entirely and work using blockchain, even in their government office.
- Along with the above organizations**, leading tech companies** like Google, Microsoft, Amazon, IBM, Facebook, TCS, Oracle, Samsung, NVIDIA, Accenture, PayPal, are working on Blockchain extensively.

# Project ideas with Blockchain technology (1/2)

- **Cryptocurrency Wallet**: Create a simple cryptocurrency wallet application that allows users to send and receive digital assets.
- **Blockchain Explorer**: Develop a web-based application that allows users to view and search the transactions on a specific blockchain.
- **Smart Contract**: Implement a simple smart contract on the Ethereum blockchain that can be used to manage a digital token or asset.
- **Voting System**: Create a blockchain-based voting system that allows for secure and transparent voting while maintaining voter anonymity.

# Project ideas with Blockchain technology (2/2)

- **Supply Chain Management**: Develop a blockchain-based system for tracking the movement of goods and services through a supply chain, providing greater transparency and traceability.
- **Decentralized marketplace**: Create a decentralized marketplace using blockchain technology where the goods and services can be directly bought by the customers without any intermediary.
- **Identity Management**: Create a decentralized digital identity management system that allows users to control their personal information and share it securely with others.

# Blockchain Wallet

- A blockchain wallet is a software that enables sending and receiving cryptocurrencies such as Bitcoin, Ethereum, etc.
- It stores the record of transactions and public and private keys which are used to perform transactions.

# Blockchain Wallet

- A **public key** is like an account number. If A wants to send some money to B using Bitcoin when A sends the public key address to B. Anyone can send Bitcoin using the public key.
- A **private key** is like an account password. Only the account holder knows the private key. The private key is used to send money

# What is Blockchain Wallet?

- A blockchain wallet is a cryptocurrency wallet that is used to manage cryptocurrencies like Bitcoin and Ethereum.
- It helps to exchange funds easily and the transactions are more secure as they are cryptographically signed.
- The privacy and the identity of users are maintained, and it provides all the features that are necessary for secure and safe transfers and exchange of cryptocurrencies.

# Features Of Blockchain Wallet

- **Secure**: Blockchain wallets are secure as they carry on the private key that needs to be secured. There are no cryptocurrencies in blockchain wallets.
- **Easy to Use**: Blockchain wallets are easy to use as they are just like any other digital wallets like Paytm that everyone uses in day-to-day life.
- **Low transaction fees**: The cost of transferring funds using blockchain wallets is much lower than that of traditional bank transactions.
- **Instant transactions**: Blockchain wallets allow users to instantly transfer funds across different geographics and these are barrier-free and without intermediaries.
- **Allows multiple cryptocurrencies**: Blockchain wallets allow transactions across multiple cryptocurrencies and thus helping users to do easy currency conversions.

# Criteria To Choose Blockchain Wallet

- There is a tradeoff between convenience and security.
- Convenient wallets are less secure while secure wallets are not as convenient to use.
- If the funds are used for a shorter time and easily accessible for day-to-day transactions, then choosing hot wallets while cold wallets will be helpful if considering storing funds for a longer period.
- Choose a blockchain wallet that allows one to control the private keys and save them on the local device or offline.
- Choose a blockchain wallet compatible with multiple operating systems.
- Choose a blockchain wallet that comes with a backup seed phrase and with security features like passwords.
- Select a wallet that is easy to use and has an active development company for maintenance, support, and improvement.

# Benefits of Blockchain Wallets

- **No geographic barriers**: Blockchain wallets helps to facilitate borderless transactions across different geographic locations without difficulties of conversions and expenses of foreign exchange.
- **No intermediaries**: There are no intermediaries in transactions carried out using blockchain wallets. Thus, there is no single point of failure as there is a central governing authority.
- **Fewer transaction fees**: There are fewer transaction costs using blockchain wallets, especially when transacting huge amounts of money.
- **Security**: Blockchain wallets provide better security and privacy of transactions due to the use of cryptography.
- **Simple signups**: Blockchain wallets have simple sign-up procedures compared to bank accounts that have strict legal and complicated procedures and verification needs.
- Easy to manage: They are easy to create, and manage.

# Challenges of Using Blockchain Wallets

- **Low acceptance**: They have low acceptance and applications as there are still many retailers and merchants that don't accept cryptocurrencies.
- **Price fluctuation**: The volatility of cryptocurrencies is one major cause of the less popularity of blockchain wallets.
- **Limited network**: Blockchain wallets have limited support for legacy and the network is also limited.
- **Complicated Refunds**: If the cryptocurrency is transferred mistakenly into some person's account, then there is no way to get a refund of the amount paid. All one can do is to ask the person for a refund and if the request is turned down, then just forget about the money.
- **Securing seed phrase**: The security of the account may get compromised if the seed phrase is not kept secure.

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# Q&A

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